
Autonomous Intelligence as a Control Architecture

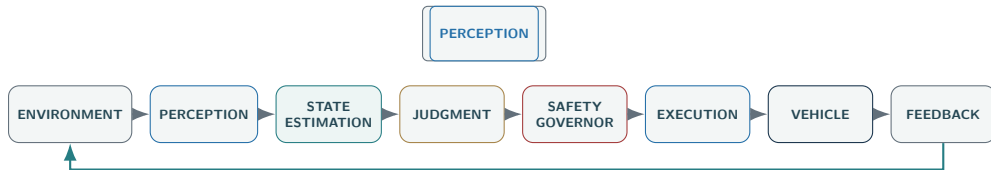
The Self-Driving Car Case

From perception to governed action in a 25-cell simulator

Autonomous intelligence is not an LLM. It is a closed-loop architecture of complementary components.

The central proposition: the car is driven by an architecture

The moving vehicle is the consequence of a recurrent chain of observation, estimation, judgment, governance, execution and feedback — not of one model acting alone.



Intelligence

Different components solve different problems: sensing, filtering, physical calculation, contextual interpretation and constraint enforcement.

Authority

Capability and authority are separate. Claude may recommend; validation and the safety governor determine what may reach execution.

Cells 1–8: build reality before asking the system to reason

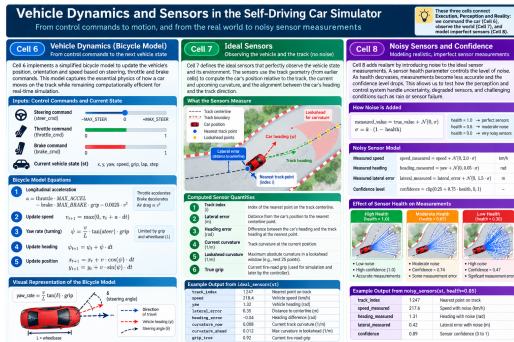
A dependable autonomous system needs an explicit world model, physical state, dynamics, sensors and uncertainty before higher-level judgment becomes meaningful.

Cells 1–5 · World and geometry

- ▶ Reproducible configuration and physical limits
- ▶ Smooth circuit centerline, heading and curvature
- ▶ Explicit vehicle state: position, yaw, speed, grip, lap
- ▶ Ground truth kept separate from what the controller observes

Cells 6–8 · Reality and sensing

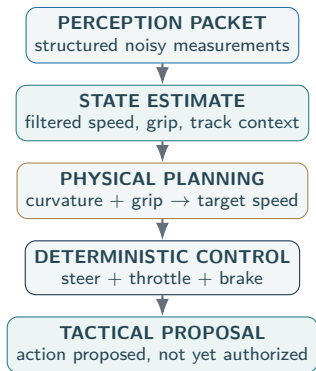
- ▶ Bicycle dynamics convert commands into motion



The notebook deliberately separates physical state, observation and confidence.

Cells 9–14: from noisy evidence to fast deterministic control

The fast loop converts uncertain measurements into a stable state estimate, a physically informed target and bounded steering, throttle and brake proposals.



Planning is constrained by physics

$$v_{\text{safe}} \approx \sqrt{\frac{a_y}{|\kappa|}}$$

Curvature κ , estimated grip and braking margin determine a forward-looking speed envelope. The language model is not required to calculate it.

Why deterministic control?

- ▶ narrow objective and short time horizon
- ▶ predictable timing and clipping
- ▶ transparent response to path and speed error

Cells 15–16: Claude contributes judgment — not actuator authority

Claude Haiku 4.5 acts as a slower strategic supervisor: it interprets context and recommends a bounded policy that must be validated before use.



Strategic contribution

- ▶ Integrates context across conditions
- ▶ Chooses NORMAL, CAUTIOUS or EMERGENCY
- ▶ Adjusts policy parameters, not steering directly
- ▶ Operates at a slower cadence than control

Validation boundary

- ▶ Unknown modes → conservative default
- ▶ Speed factor clipped to [0.45, 1.05]
- ▶ Braking margin clipped to [1.0, 1.6]
- ▶ Types normalized; rationale bounded

Generated output is treated as untrusted structured input

Cells 17–20: governance turns proposals into authorized action

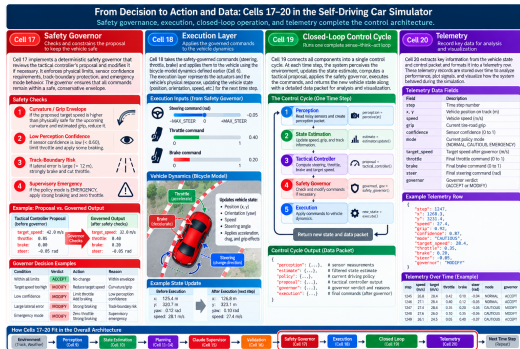
The safety governor is the decisive authority boundary: it evaluates the current state and can modify any otherwise valid tactical proposal before execution.

Four explicit safety checks

1. **Curvature / grip envelope:** reduce excessive target speed
2. **Low confidence:** limit throttle and add braking
3. **Track-boundary risk:** strong braking for large lateral error
4. **Emergency mode:** zero throttle and impose braking

Then the chain closes

- Execution applies only the governed command



Cells 21–25: stress the architecture, not just the model

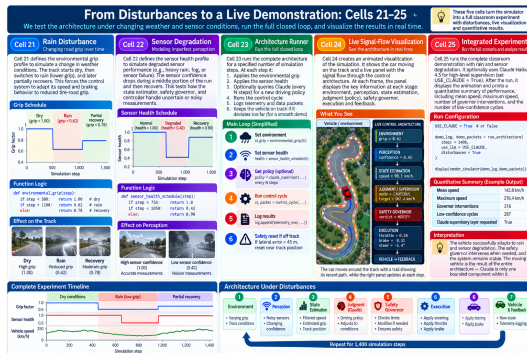
The integrated experiment separates physical difficulty from uncertainty about the physical state and observes whether the whole system adapts coherently.

Disturbance schedule

- ▶ Grip: 1.00 $\rightarrow$ 0.62 $\rightarrow$ 0.78
- ▶ Sensor health: 1.00 $\rightarrow$ 0.42 $\rightarrow$ 0.90
- ▶ Rain and sensor degradation overlap only partly
- ▶ 1,400 repeated simulation steps

What should become visible?

- ▶ lower grip $\rightarrow$ lower safe target speed
- ▶ lower confidence $\rightarrow$ independent caution
- ▶ combined stress $\rightarrow$ stronger constraint
- ▶ recovery $\rightarrow$ gradual return toward



Autonomy is a hierarchy of timescales and authorities

Different layers should operate at the cadence appropriate to their function and possess only the authority required by their role.

Layer	Primary function	Cadence	Authority
Perception	Measure the world	Fast	Observation
State estimation	Construct operational state	Fast	Inference
Planning / control	Compute local action	Fast	Proposal
Claude supervisor	Interpret context; set policy	Slower	Recommendation
Validation	Normalize model output	Supervisor call	Admissibility
Safety governor	Enforce hard constraints	Fast	Modify / constrain
Execution	Apply approved commands	Fast	Physical action
Telemetry	Preserve causal evidence	Fast	Audit

Timescale separation

API latency or strategic reasoning must never define steering-loop stability. Fast control continues under the most recently validated policy.

Authority separation

Broad model capability does not imply broad permission. Architectural position determines practical power.

What the simulator teaches about failure and resilience

Reliability comes from locating where assumptions can fail, assigning each failure a detection signal, a bounded response and a durable record.

Failure can originate in different layers

- ▶ **Perception:** noisy or degraded sensors
- ▶ **Estimation:** lag or poor filtering
- ▶ **Control:** oscillation, saturation, delayed braking
- ▶ **Judgment:** valid but strategically poor policy
- ▶ **Interface:** units, schema, stale messages
- ▶ **Evidence:** incomplete or unjoinable telemetry

The architecture contains failure

- ▶ Confidence propagates uncertainty
- ▶ Validation catches malformed semantic output
- ▶ Governor enforces invariants every fast cycle
- ▶ Deterministic control survives supervisor unavailability
- ▶ Proposal and execution remain separately logged
- ▶ Recovery is evaluated, not only degradation

A robust autonomous system is not one that never fails — it

From the self-driving car to autonomous financial institutions

The architectural lesson generalizes: dependable autonomy combines specialized intelligence, bounded permissions, independent controls and evidence.

Self-driving car	Financial institution
Sensors	Market data, ledgers, documents, client inputs
State estimator	Risk, liquidity and exposure state
Curvature / grip constraints	Capital, liquidity, legal and risk limits
Tactical controller	Pricing, hedging, routing, portfolio logic
Claude supervisor	Contextual judgment, exception analysis, strategic synthesis
Safety governor	Compliance gates, approval matrices, hard limits
Actuators	Orders, payments, approvals, communications
Telemetry	Audit trail, model logs, control evidence

Autonomous intelligence is not “Claude driving.”

It is a governed closed loop in which Claude is one bounded source of judgment.

Perception → State → Judgment → Validation → Governance → Execution → Feedback